

CS1060 Assignment 2: Group Travel Planner

Peter Chen

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1 Product Naming

The tentative name for this product is *Covalent*. This name symbolizes that this product creates and strengthens bonds.

2 Product Description

Covalent is a collaborative trip planning platform that has the following features:

- Group coordination: In-app group chats for trip planning with features like polls, suggestions, shared wishlists, etc.
- Automated aggregation: AI organizes group members' ideas into detailed itineraries, with activities, accomodation, transport, food options, tickets. The itineraries could be individualized and even differ for members of the same group, given travel restrictions, time conflicts, activities preferences, etc.
- Budgeting & Booking: Generates an overall budget breakdown and integrates booking links for seamless purchasing. If purchased in bulk given large groups, there may also be potential discounts through partnerships.
- Social discovery: The app also connects users with other travelers in the same area, with personality/preference matching for shared activities.
- Shared memories: The app logs the footprint of everywhere the user traveled to, and the groups and people the user is with at each location. There can be shared photo albums added through the app for each trip.

3 Similar Services & Differentiation

Existing Services

1. **TripIt / Wanderlog**: Both of these apps focus on itinerary building and syncing bookings, but not collaborative ideation or social matching.
2. **Hopper / Kayak / Expedia**: Flight, transportation, deals bookings and planning.
3. **Travefy / Triptile**: Itinerary builder for travel professionals and group organizers.
4. **Messages / Facebook / Whatsapp**: widely used for trip planning, but manual, messy, no automation, and no booking integration.
5. **Backpackr / Travello / Tinder / Hinge**: Social discovery
6. **Google Photos**: Shared photo album

Differentiation

- **Integration of planning and social platform.** Users can go from brainstorming a trip with friends, to develop itineraries and bookings, to inviting other like-minded travelers to join, to sharing experiences with the broader community.
- **AI-driven personalization and itinerary aggregation.** The app leverages AI as a core functionality to simplify travel planning, including a continuous chat box to input ideas and generation of personality-based recommendations. AI will also aggregate and learn from community data to improve itineraries.
- **Prioritize user experience.** The app will have a clean interface, straightforward controls, and almost gamified component to itinerary generation, with features like drag-and-drop itinerary building, visual map-based planning, shareable trip summary page, and global travel footprint of users that can be shared.
- **Socialization and community.** Users that access social aspect of the app will require verification to ensure safety when meeting up. Traveling is inherently a very socializing activity, and this app rides on that aspect.

4 Stakeholders

Customers

- Demographic: both individual consumers (trip organizers, members that would pay for premium features) and organizational clients (companies).
- Values: On the individual side, planners or friend groups wish to become organized and figure out a detailed itinerary for the trips. Planners might wish for everyone to use the tool and participate, so the app should be easy for invitees to engage. For corporate customers, the app needs to understand both the company's objectives (cost, retreat) and the employees desire for fun and relaxation. Companies would also value data security.
- Motivations: Young adults wish for a easy, fun way to coordinate a trip with their friends and are motivated by planning a trip that everyone enjoys. A tool that can save money and keep everyone informed is a bonus. Corporations are motivated by efficiency, outcome, and reliability.
- Expected behaviors: Individuals might start using the app for a specific trip idea, and if it works well, become repeat users. They are likely to invite their friends into the app, driving network effects. Corporate might require onboarding help or custom features.

Users

- Demographic: all end users of the app, including friends invited to a trip plan, employees attending the retreat, and solo travelers or small groups using the app to meet others.
- Values: There are a few different core values that arise for users, including planners vs. spontaneous travelers, privacy vs. social sharing, work vs. leisure. The app should be straightforward and transparent within the travel group.
- Motivations: everyone that uses this app wishes to have a memorable trip with minimal hassle, and usually low cost. They're motivated by cool activities and interesting people to meet.

- Expected behaviors: log in to check latest plan updates, chat in the group, and refer to itinerary or get notifications during the trip. Browse community to find like-minded travelers.

Employees:

- Demographic: product developers, designers, customer support, sales. People building and maintaining the software.
- Values: building the app with the ideal user experience while not burning too much cash to bankrupt the company. As a startup, employees with equities are motivated to work more, but work-life balance is important.
- Motivations: mission alignment with the company, career growth opportunities, user satisfaction, and innovation.
- Expected behavior: agile development through the early stages, monitor user analytics. Sales team interacts with travel partners or potential B2B clients.

Investors:

- Demographic: angel investors, venture capitalists, founders/employees with equity.
- Values: investors value the return that they are receiving on their investments. Value-driven investors also care about the values and mission of the company.
- Motivations: they are primarily motivated by return on investment, market disruption, and various metrics to value the company.
- Expected behavior: investors may care about the returns or the company enough to take a seat on the board or have regular check-ins. They might connect with potential partners or hires, or apply pressure if the targets aren't met.

Owners:

- Demographic: Me and my co-founders that I find later down the line.
- Values: we're primarily driven by the idea and the monetization value it brings.
- Motivation: to put it bluntly, I would love to build a company into a unicorn.
- Expected behavior: continuously talk about the product, find ways of improvement, fast development, fundraising.

Regulators:

- Demographic: Law enforcement, legislature.
- Values: they value safety and legality of each aspect and feature of the product.
- Motivation: they are driven by security, safety, and will to protect their citizens.
- Expected behavior: check in on security of the app

Business Partners:

- Demographic: hotels, airlines, transportation companies, clubs
- Values: they are driven by the monetization provided from the partnership.
- Motivation: they are motivated by money and customer acquisition.
- Expected behavior: partner if they believe my product can bring them value.

5 User Journey

Sample primary user: “Alex”, who has a friend group of 5 in their mid-20s planning a one-week overseas vacation together. Alex is the initiator to organize the trip using our app.

1. Discovery:

- Alex sees a TikTok demo showing group votes on destinations, and recalls last year’s headache juggling chat threads + Google Sheets.
- Downloads the app, tries an interactive demo trip.
- Value tension: “Is this better than what we already use?” → Smooth demo + social proof (e.g., “500k trips planned here”) convinces Alex to invite friends.

2. Onboarding + Trip Setup:

- Alex creates “Thailand 2026” trip, invites 4 friends by share link. No heavy signup; even Bob (tech-averse) joins via browser link and taps date preferences.
- The app auto-creates a group chat + calendar poll.
- A playful quiz (“Beach party or cultural tour?”) seeds AI ideas.

- Example: The app suggests “late May” since all 5 marked that week green on the poll.

3. Collaborative Ideation:

- AI drafts 3 destination/itinerary options (Thailand, Bali, Mexico) with budget ranges: “Thailand $\approx$ \$800/person, 7 days, mix of beaches + temples.”
- Friends comment under each; emoji reactions show interest.
- Timeline view lets them drop in personal wishes: one adds Elephant Sanctuary, another adds Bangkok nightlife.
- Value tension: Tight budget vs. big ambitions $\rightarrow$ Running cost per person shows \$750 vs. \$1,200 plans, making trade-offs visible.

4. Booking & Payments:

- They settle on Thailand. Alex finds flights; each friend taps “Confirm & Pay my share” within 24-hr soft hold.
- Hotel chosen via search; receipts + PDFs auto-added to itinerary.
- Example: Expense tab shows Alice paid deposit; Bob owes \$60 $\rightarrow$ auto reminder goes out.
- Value tension: Quick action vs. group consensus $\rightarrow$ app shows “2 seats left at this fare” and highlights poll majority, nudging timely booking.

5. Trip Execution:

- Each morning: push “Today: Island Tour at 10 AM, bus pickup at lobby.”
- Rain forecast $\rightarrow$ AI suggests Thai cooking class as alternate; group votes yes.
- They use “Travelers Nearby” to see another group in Bangkok, join them for nightlife. Safety net: profiles show past reviews, and live location share reassures cautious members.
- Value tension: Structured plan vs. spontaneity $\rightarrow$ Itinerary blocks marked “optional/free” so free-spirited friends can peel off.

6. Post-trip:

- App compiles expenses: “Alice owes Bob \$45, Charlie owes Dana \$30” $\rightarrow$ one-tap Venmo settles.

- Trip highlights auto-generated: “7 days, 3 cities, 50 meals shared.” Friends post photos, some make their itinerary public.
- Quick 3-question survey (“What did you love most?”) trains AI for future trips.
- Value tension: Privacy vs. sharing → By default album stays private; Alex opts to share highlight reel to community feed.